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P-XX

Two-Layer Topological Mapping for Energy-Aware Arm and Base Selection on a Dual-Gantry Quad-Arm Ceiling Robot

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One topological representation, two layers: the room the robot sees, and the reach each arm owns, priced in energy.

Background

- Aging population → assistive robots **in-home**
- Small, cluttered, **shared with residents**
- **Ceiling-mounted** → floor stays free
- **2 rails** × **2 arms** = 4 arms, **8-DOF** each
- **RGB-D only**, no LIDAR

Problems

- **P1:** which arm, where the base stops — coupled
- **P2:** fixed-base maps only, reachable / not
- **P3:** occupancy has no topology

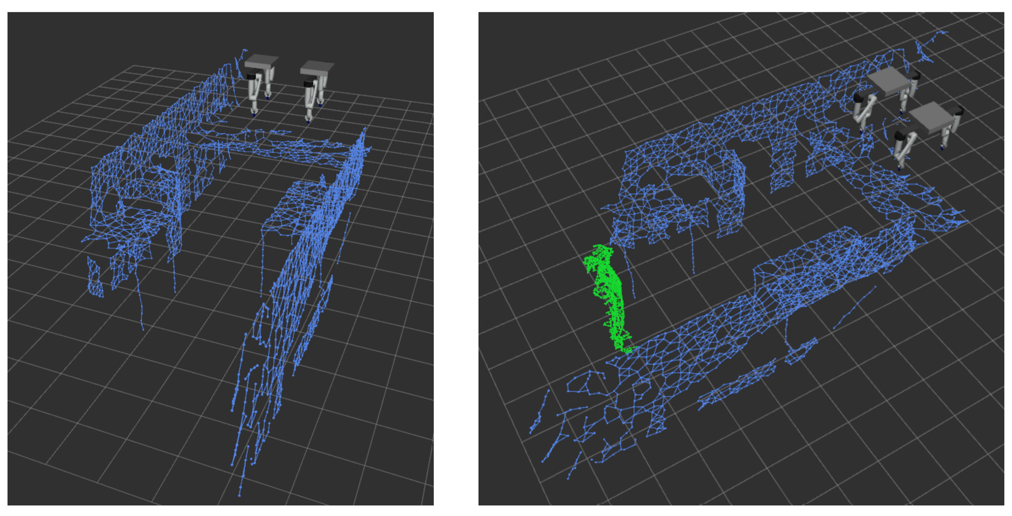
Core Idea: One GNG, Two Layers

One Growing Neural Gas, fitted twice: what the cameras see (environment) and what each arm can reach (capability).

Layer 1: Environment GNG from RGB-D

- 2 overhead RGB-D cameras, **no LIDAR**
- **Placement:** ceiling-mounted ($z \approx 2.05$ m), diagonally opposite corners, angled down/inward at the table
- `depth_cloud`: depth → world points, no seg. gate
- GNG on the cloud → scene *topology*
- **Static:** frozen once. **Dynamic:** live remainder only
- Union → MoveIt 2 planning scene; tracks by Hungarian match

*The two overhead RGB-D views: the entire input (**NVIDIA Isaac Sim**).*

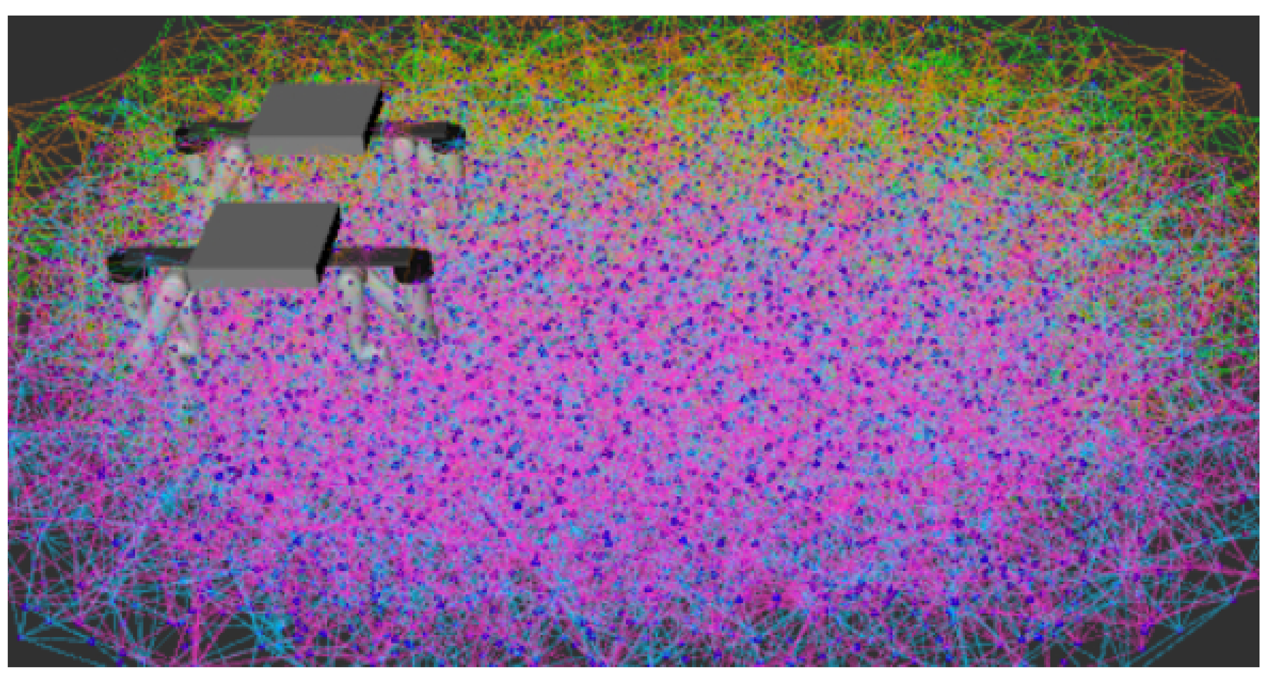


Left: frozen static backbone. Right: plus dynamic remainder (a person).

Static vs. dynamic budget: 1800-node static backbone (offline, multi-epoch, captured once) vs. 800-node live cap (online, single pass per frame) — same GNG, different node budget & training regime per layer.

Layer 2: Base-Aware Capability GNG

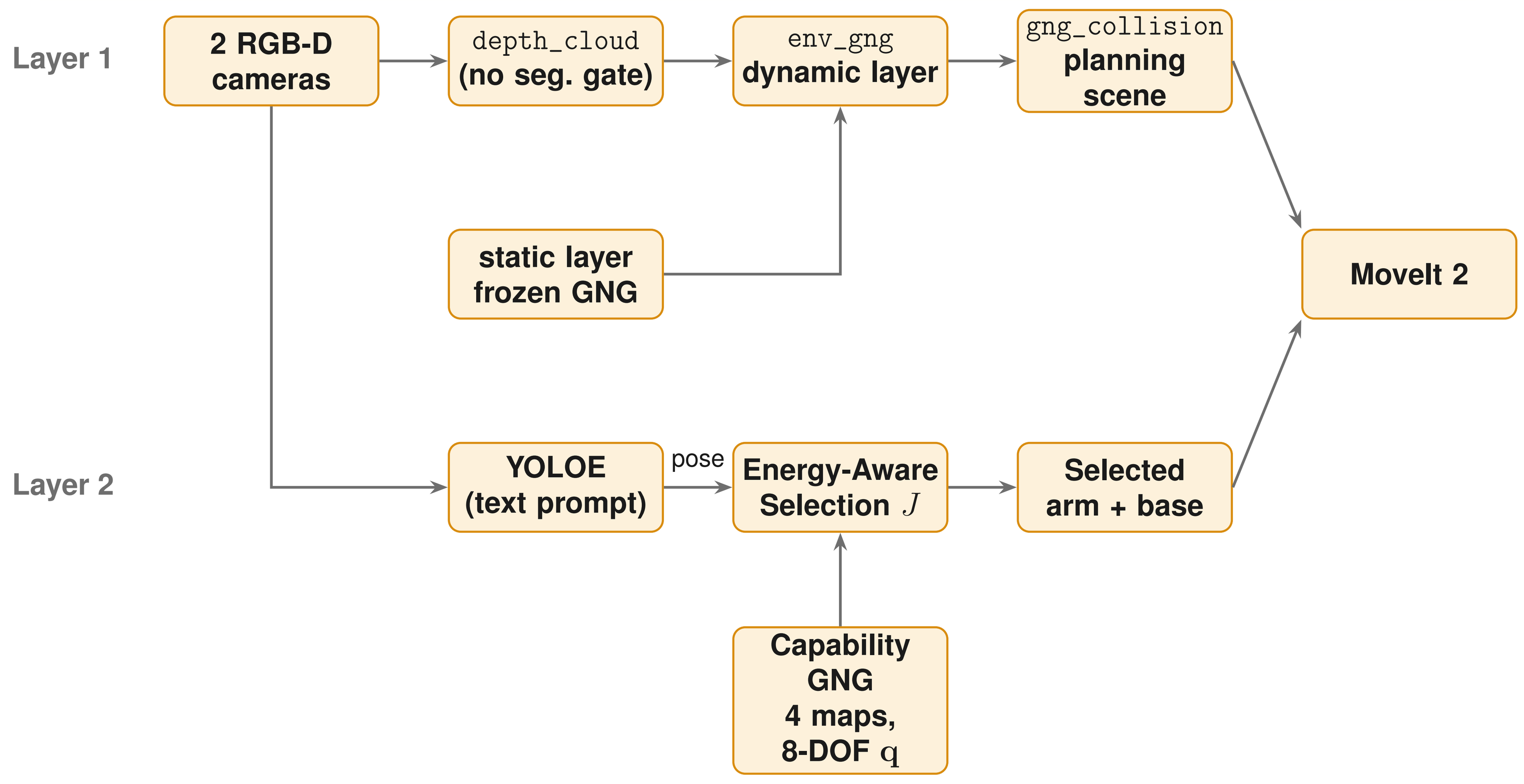
- Sample **full 8-DOF** $\mathbf{q}$, FK (Pinocchio), manipulability $w = \sqrt{\det(JJ^\top)}$
- Node stores $[\mathbf{x} \mid \mathbf{q}]$: point + config, **rail included**
- **Boundary seeding** → 4 maps (one per arm)



Capability layer of one arm (RViz), coloured by manipulability.

Per arm: 80,000 FK samples → **2915 nodes** (600 boundary), 14,547 edges, spacing **0.090 m**.

System and Pipeline



Layer 1 (geometry) is decoupled from Layer 2 (identity + decision).

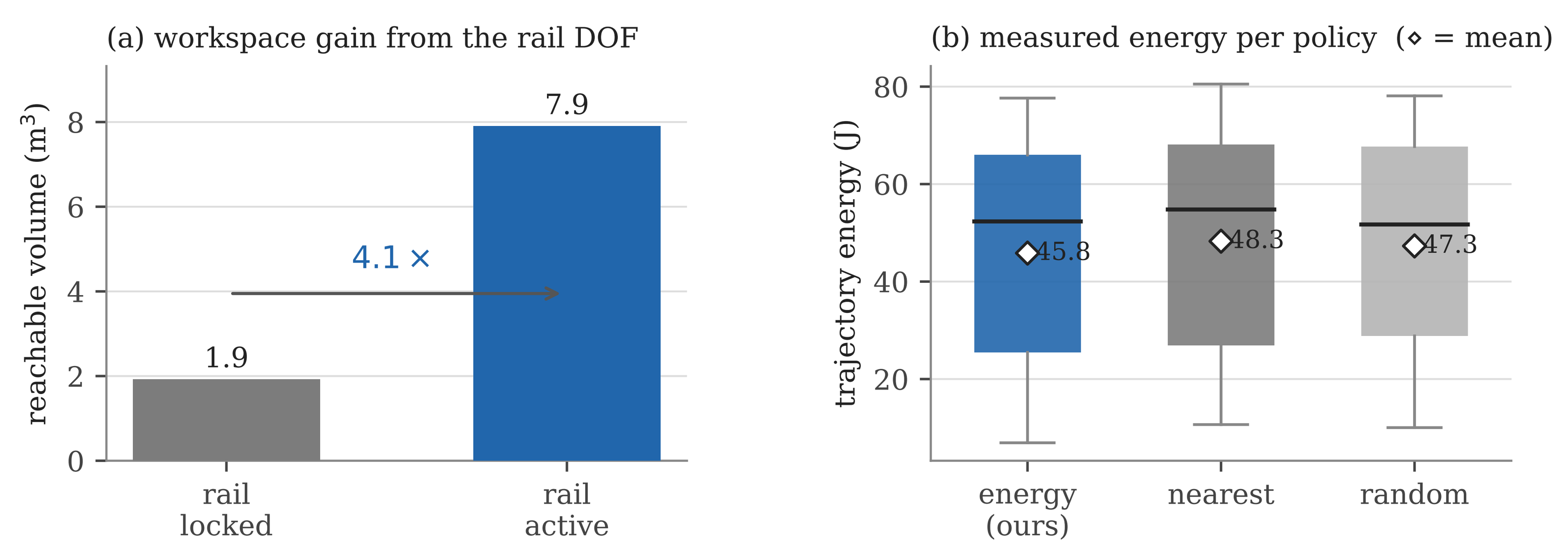
Energy-Aware Selection Cost J

$$J = w_{gl}\frac{\Delta_{lin}}{\rho_{gl}} + w_{gr}\frac{\Delta_{rot}}{\rho_{gr}} + w_{arm}\frac{\Delta_{arm}}{\rho_{arm}} + w_{dist}\frac{e}{\rho_{dist}} - w_{manip}\frac{m}{\rho_{manip}}$$

- Rail/arm travel, tool distance, manipulability — **fit by OLS** on 168 picks ($R^2 = 0.857$): rail dominates
- **GNG** → J: every **capability-layer node** near the target (all 4 arms) scored by J from its own $\mathbf{q}$
- **Selection: lowest-J first**, **plan-only**; first collision-free plan wins

Experiments and Results

- **Feasibility, not just efficiency:** 93.3% multi-arm success vs. 63–65% for any single fixed arm
- **Why:** rail active → $4\times$ reachable volume (**1.9→7.9 m**^{3})
- **Boundary seeding:** reach-surface error **0.221 m → 0.008 m**



(a) Reachable volume. (b) Trajectory energy per policy (◇=mean).

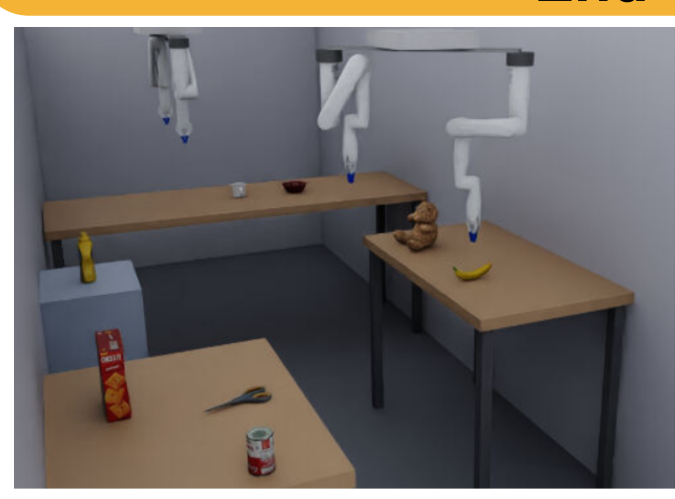
Selection policies, 60 target poses, plan-only.

Mode	Success	Rail lin. (m)	Energy (J)
energy (ours)	**56/60 (93.3%)**	**0.97**/1.01	**45.84**/52.34
nearest	56/60 (93.3%)	1.19/1.41	48.28/54.79
random	56/60 (93.3%)	1.17/1.28	47.31/51.72
fixed (best, arm_4)	39/60 (65.0%)	1.14/1.21	49.79/55.78

Fixed-arm energies use that arm's own smaller reachable set.

- **Energy (secondary):** lowest mean, but modest (**45.84 J** vs. 48.28 J, -5.1%)
- **Trade-off confirmed:** arm_3 (0.64 m rail) chosen over nearest arm_4 (1.15 m rail): rail cost avoided

End-to-End Validation



- **Full pipeline:** 35/35 (100%) picks, 39.6 s/pick avg
- **Perception:** 100% detection, 0.023 m error
- **Reachability screening:** 100% (8/8) correct
- **Environment layer:** 1800 nodes static, 800-node live cap, $\sim$**1 Hz**

Summary and Contributions

- **Feasibility over efficiency:** 93.3% vs. 63–65% success
- **Base as a genuine DOF:** $\approx 4\times$ reachable volume
- **Coupling, correctly priced:** J fit on measured energy ($R^2 = 0.857$)
- **One GNG, two layers:** one representation for collision *and* capability

Future work: real hardware; dynamic-scene study.

References. [1] Fritzke, NIPS 1995 (GNG). [2] Zacharias *et al.*, IROS 2007. [3] Makhal & Goins, IRC 2018 (Reuleaux). [4] Yoshikawa, IJRR 1985. [5] Sandakalum *et al.*, Humanoids 2022. [6] Shao *et al.*, arXiv:2403.19940 (MoMa-Pos). [7] Naik *et al.*, IROS 2024 (BaSeNet). [8] Wang *et al.*, ICCV 2025 (YOLOE). [9] Carpentier *et al.*, SII 2019 (Pinocchio). [10] Coleman *et al.*, JOSER 2014 (MoveIt).